

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Unless otherwise advised by the manufacturer, the following procedures are recommended:

Always renew the timing chain

If the timing chain is to be re-used, mark the running direction before removal

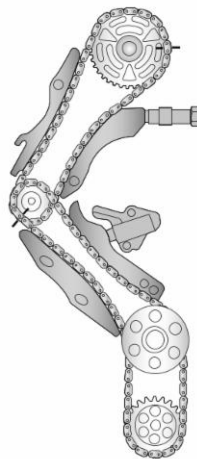
Never turn the engine in the direction opposite to that of normal operation

Note: Avoid bending, torsion or traction

Always use new gaskets

General

Timing chain layout



Removal

Disconnect the battery earth cable

Remove the engine together with the gearbox

Remove the resonator

Remove the engine acoustic cover

Remove the turbocharger air inlet pipe

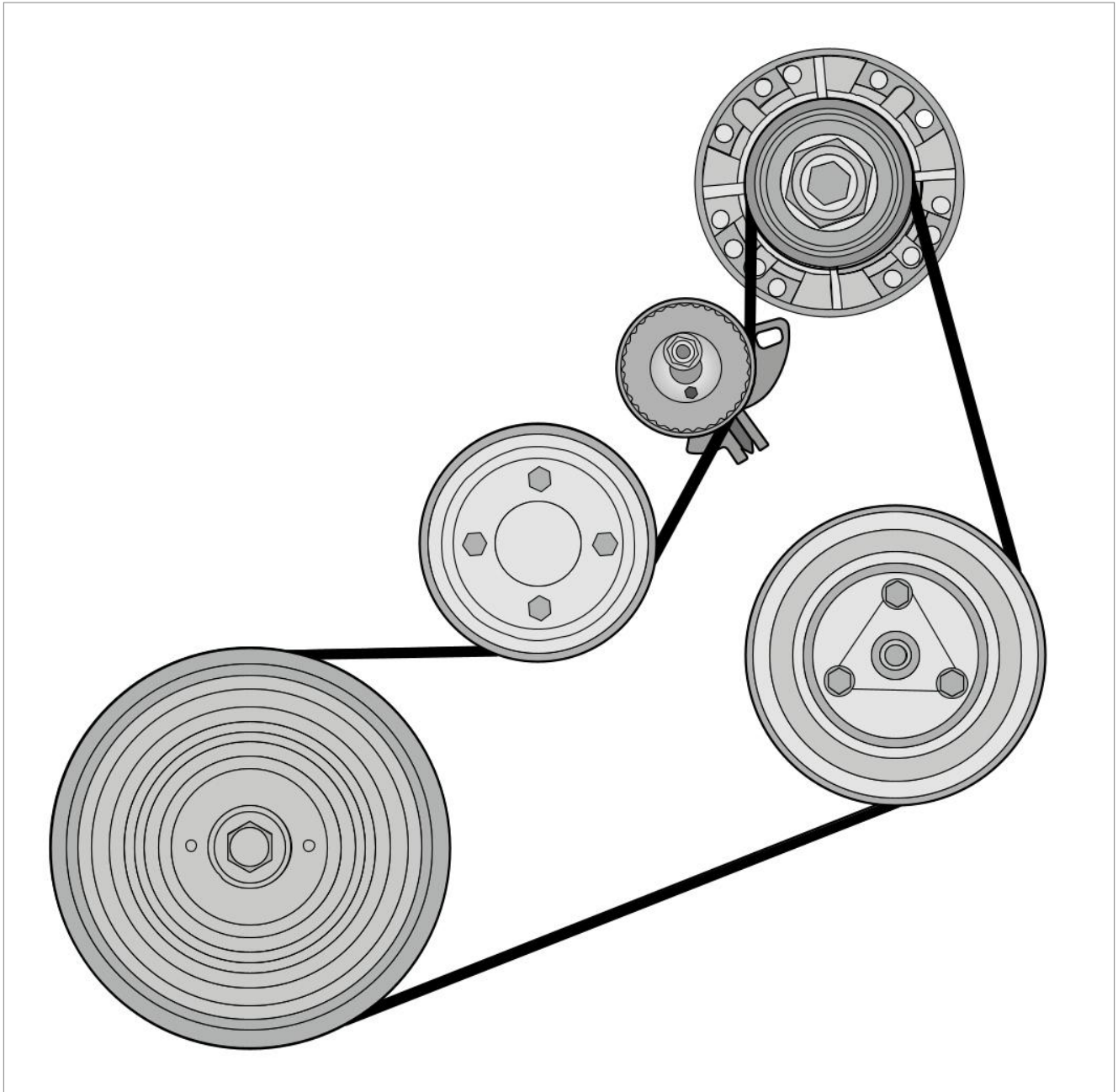
Remove the differential pressure sensor

Disconnect the engine wiring from the top of the engine

Remove the ancillary drive belt

Drive belt layout

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Remove the alternator

Remove the air-conditioning compressor

Disconnect the positive battery cable

Remove the coolant hoses

Remove the diesel particulate filter

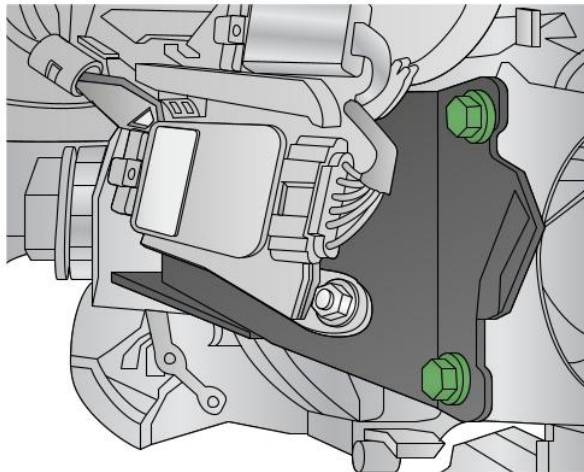
Remove the EGR cooler

Remove the turbochargers

Vehicles with 4WD:

Remove the electronic control unit support

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Remove the intermediate drive shaft

Remove the transfer box

Vehicles with automatic transmission:

Disconnect the automatic transmission oil cooler hoses from the radiator

Fit the crankshaft turning tool (11 6 480)

Slowly turn the crankshaft clockwise

Remove the torque converter attaching nuts

All vehicles:

Use a hoist to support the gearbox (11 0 020, 2 220 718)

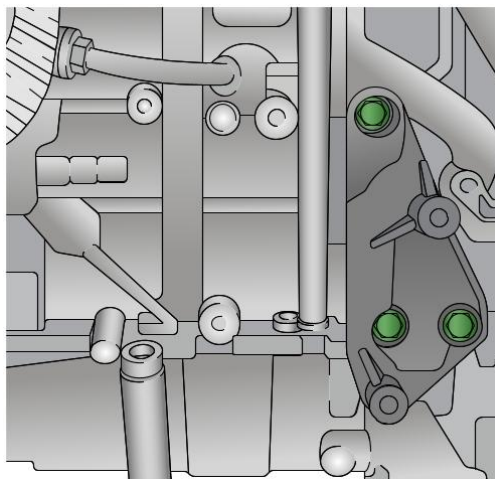
Remove the engine/gearbox attaching bolts

Separate the gearbox from the engine

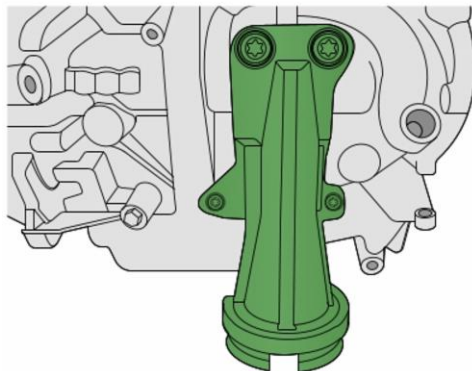
Fit a lifting eye to the cylinder head ((21.5 Nm)) (2 459 012)

Note: Lift the engine

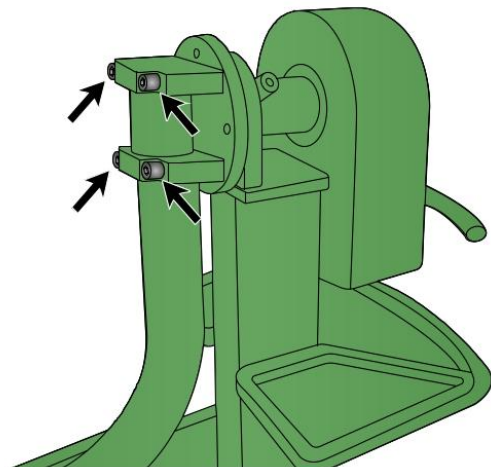
Remove the engine support bracket



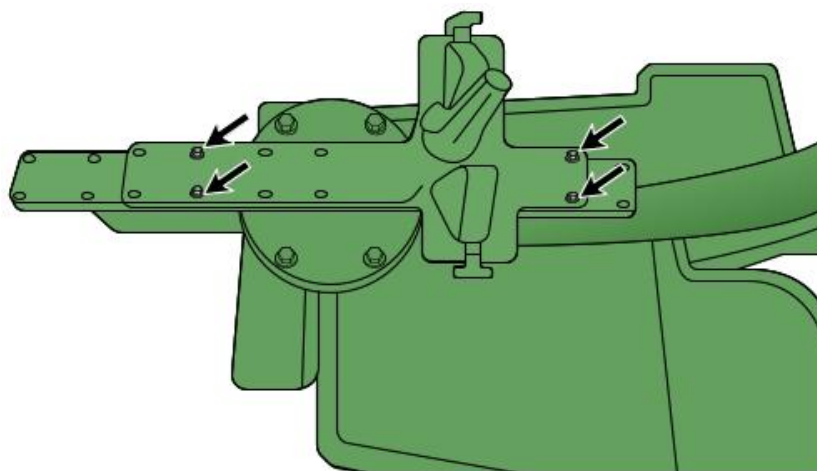
Fit the special tool ((38 Nm)) (2 455 010)

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Assemble the special tool as shown. Tighten the special tool bolts ((130 Nm)).



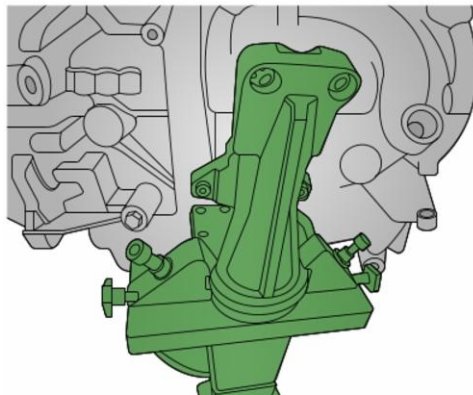
Fit the special tool ((160 Nm)) (2 413 082)



Lower the engine

Support the engine (2 413 082, 2 455 010)

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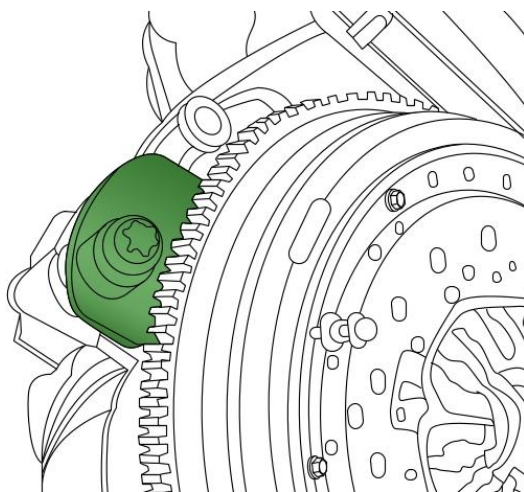


Remove the cylinder head

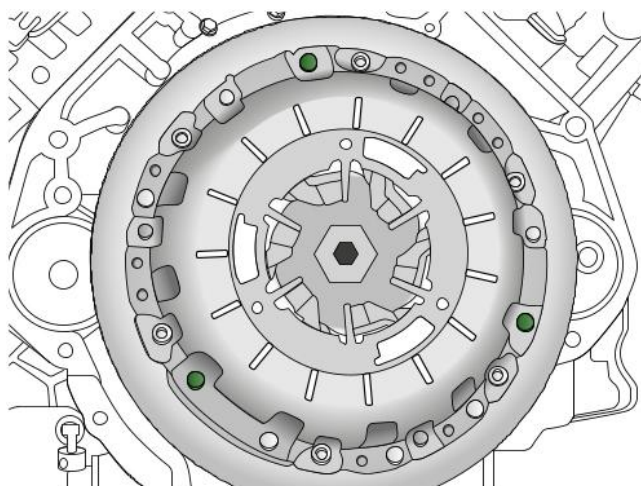
Remove the crankshaft TDC locking pin (2 288 380)

Vehicles with manual gearbox:

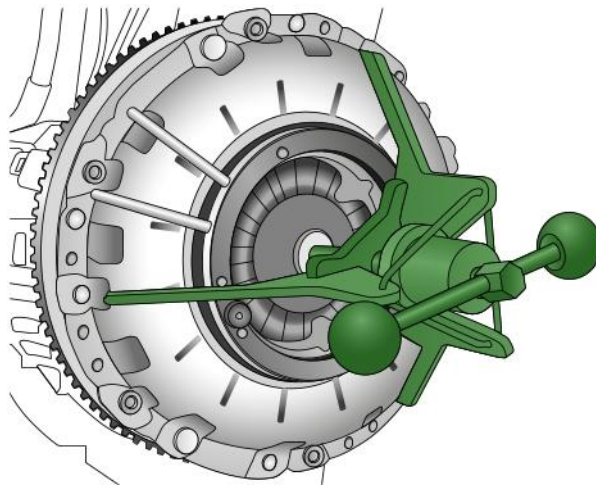
Fit the flywheel locking tool (11 9 265, 11 9 811)



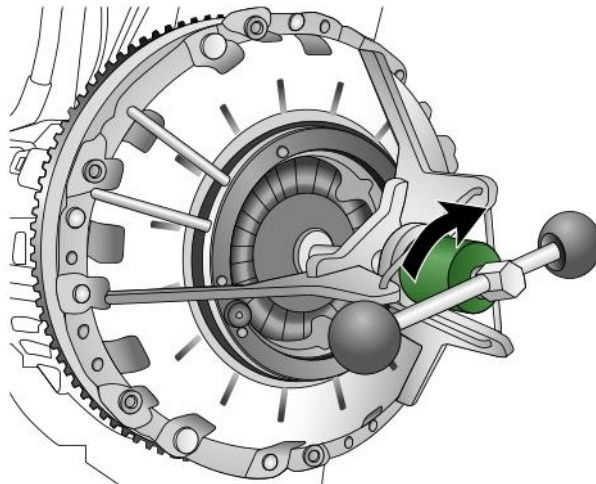
Loosen the three bolts



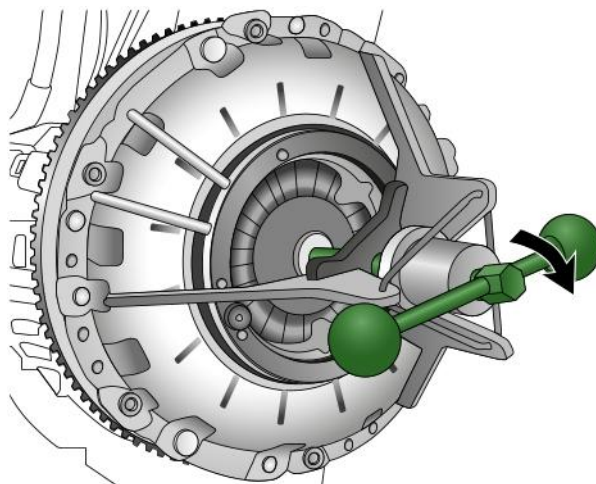
Fit the special tool (21 2 170)

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Tighten the screw as far as the stop



Tighten the diaphragm spring to its stop



Remove the clutch pressure plate bolts

Remove the clutch

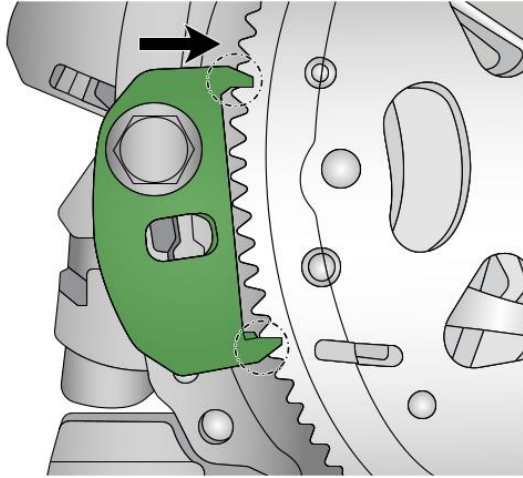
Remove the flywheel bolts. Use a socket wrench.

Remove the flywheel

Vehicles with automatic transmission:

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Fit the flexplate locking tool (11 9 260)



Remove the flexplate

All vehicles:

Remove the crankshaft position sensor gearwheel

Remove the dipstick tube

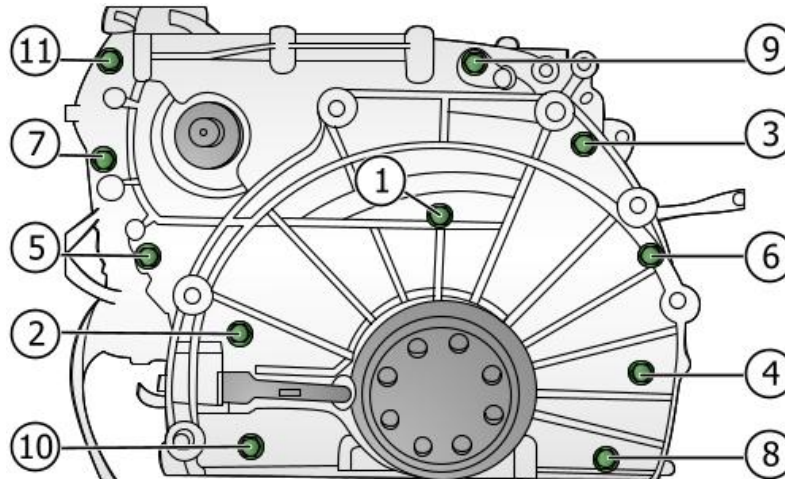
Remove the oil sump protective cover

Remove the oil sump

Disconnect the coolant hoses from the timing cover

Remove the wiring loom bracket

Remove the bolts in the reverse order to that shown ((11 -> 1))



Remove the timing chain cover. Use a screwdriver.

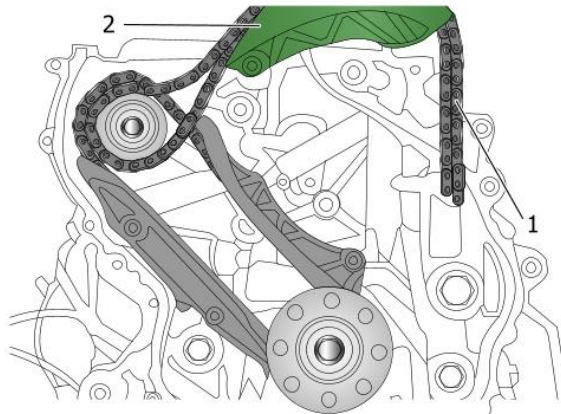
Remove the oil pump

Remove the oil pump chain

Remove the secondary timing chain ((1))

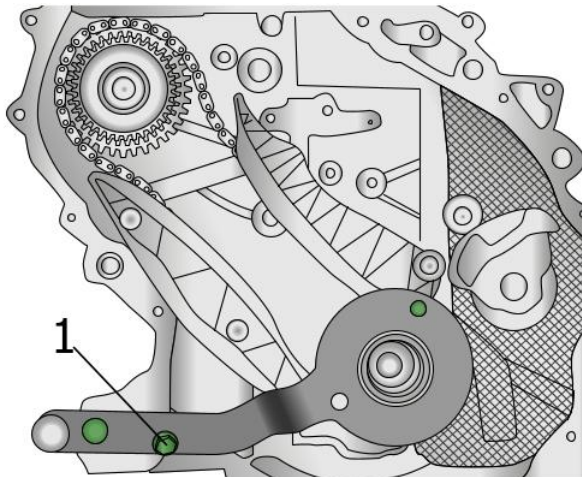
Remove the chain tensioner guide ((2))

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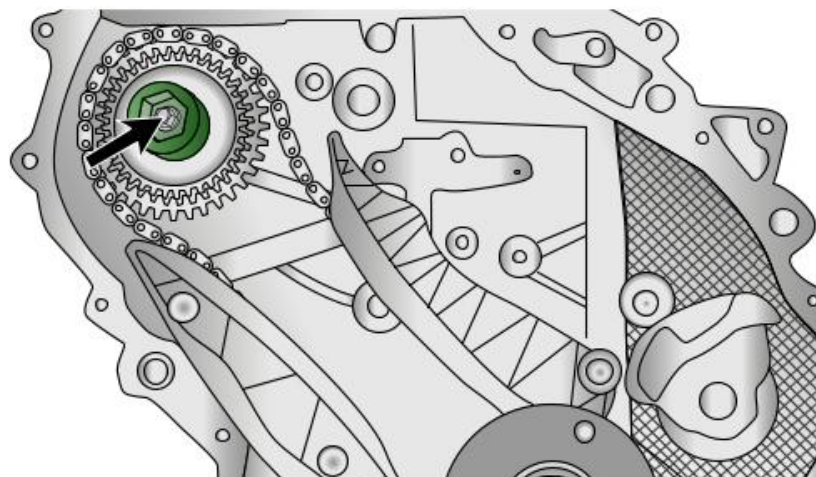
Fit the special tool at the point shown (2 357 904)

Secure the special tool with a bolt ((1))



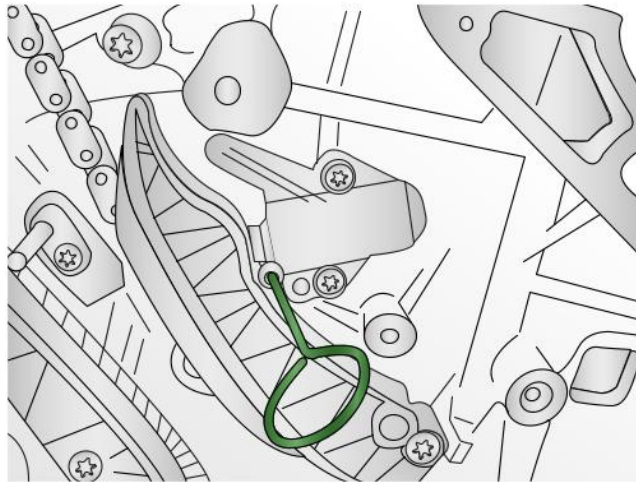
Loosen the high-pressure pump gearwheel bolt

Fit the special tool. Detach the high-pressure pump gearwheel.



Compress the primary chain tensioner

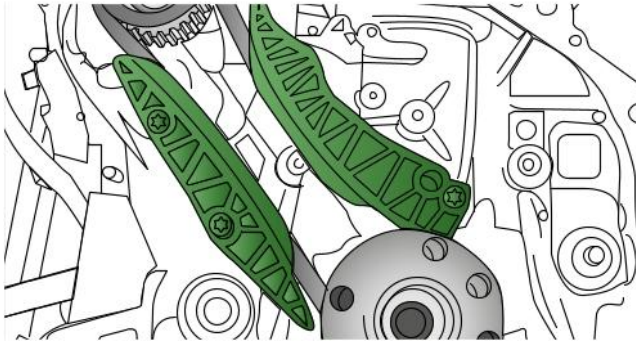
Lock the chain tensioner with a locking pin (11 4 120)

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Remove the primary chain tensioner

Remove the chain guide

Remove the chain tensioner guide



Remove the high-pressure pump gearwheel

Remove the crankshaft locking tool (2 357 904)

Remove the primary timing chain

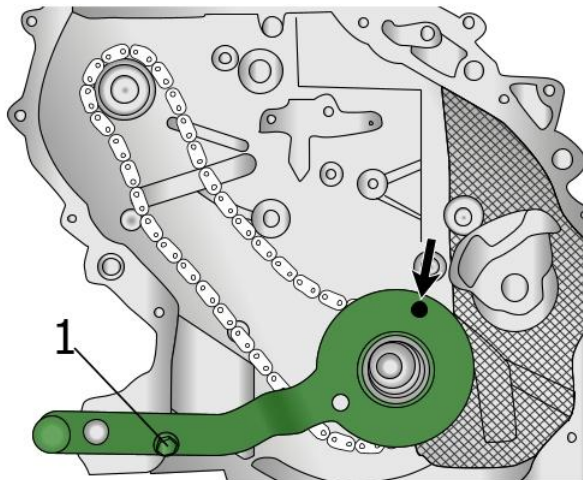
Installation

Fit the primary timing chain

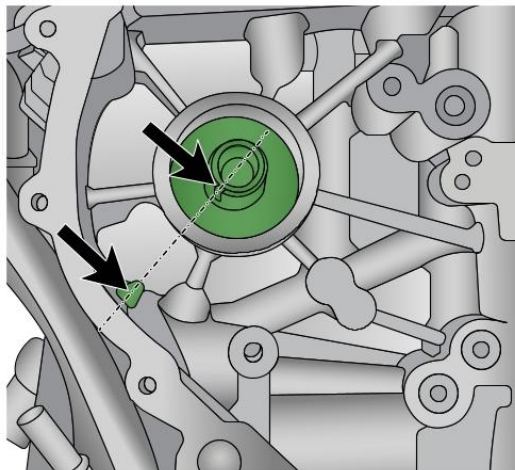
Fit the special tool at the point shown (2 357 904)

Secure the special tool with a bolt ((1))

Repair Manuals

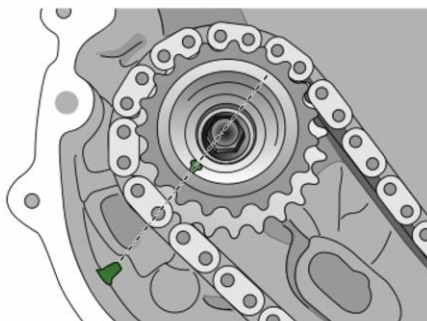


Check the timing marks



Refit the high-pressure pump gearwheel

Note: Do not tighten the bolt



Refit the chain tensioner guide

Refit the chain guide

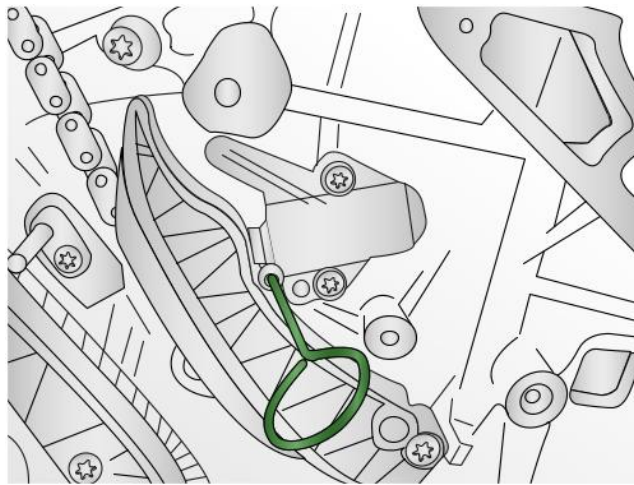
Note: Use new bolts

Refit the primary chain tensioner

Note: Clean off the sealing surfaces

Remove the tensioner locking pin (11 4 120)

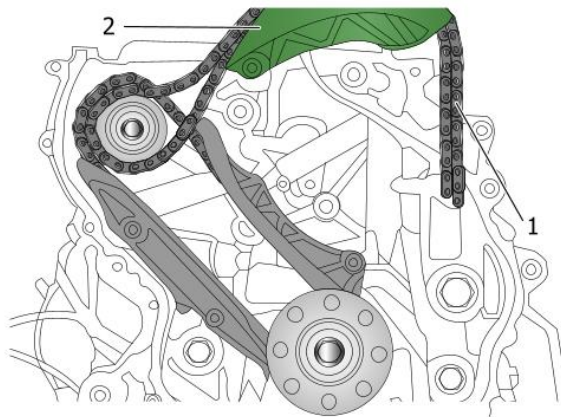
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Tighten the high-pressure pump gearwheel

Fit the secondary timing chain ((1))

Refit the chain tensioner guide ((2))



Fit the oil pump chain onto the crankshaft gearwheel

Refit the oil pump gearwheel

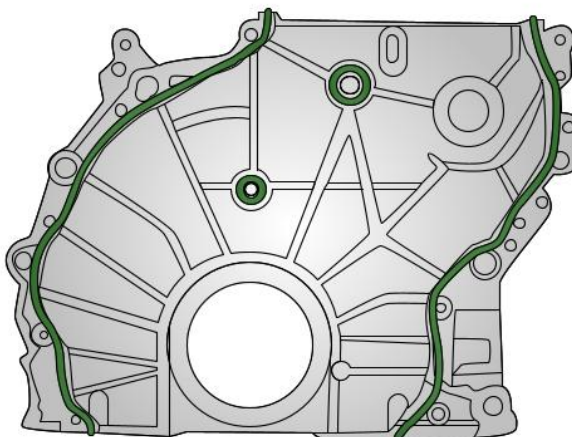
Timing chain cover:

Clean the contact surfaces

Check that the dowel pins are still in position

Apply sealant as indicated ((1.5 - 2.0 mm))

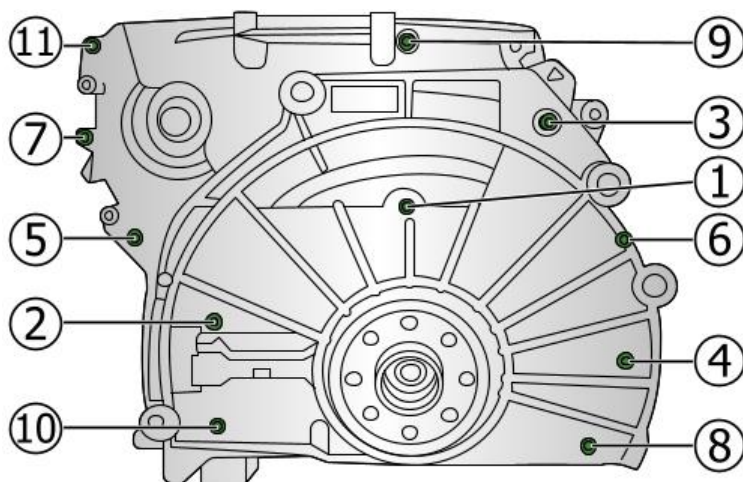
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Note: The cover must be fitted within 10 minutes of applying the sealant

Note: Use new bolts

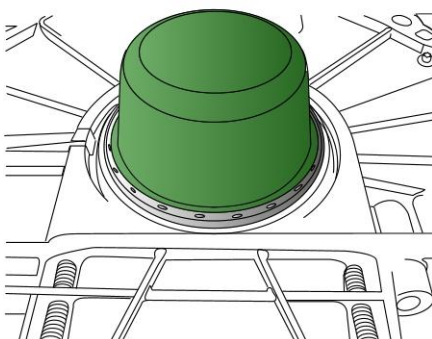
Tighten the timing cover bolts ((1 - 11))



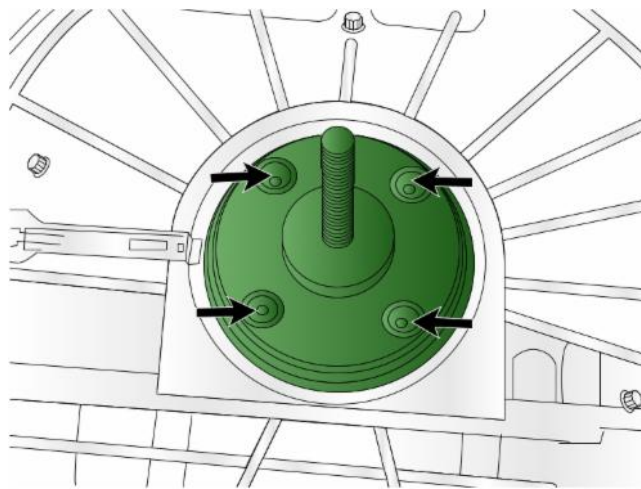
Note: Do not touch the lip of the seal

Fit the crankshaft oil seal

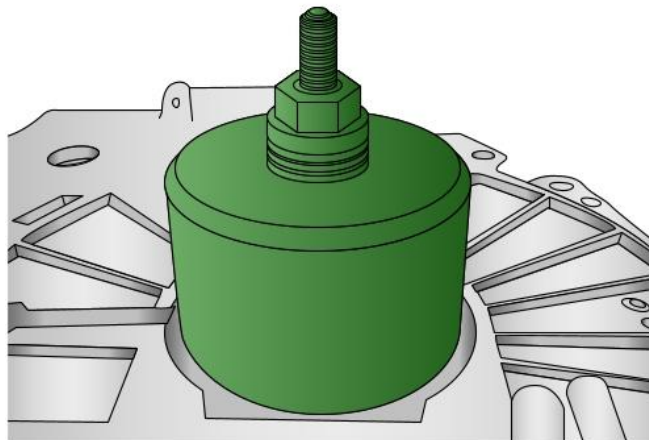
Use the special tool (11 8 815)



Fit the special tools (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)

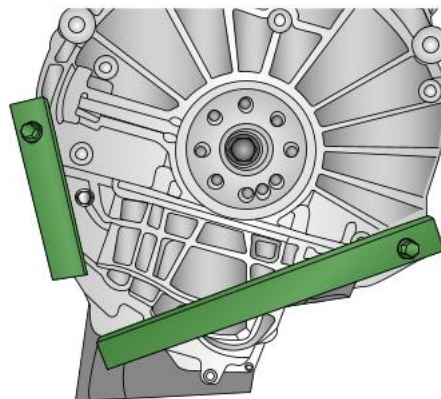
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Tighten the special tool bolt until full surface contact is made



Remove the special tools (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)

Refit the oil sump. Use the special tool.



Further refitting is in the reverse order of removal

Reconnect the battery earth cable

Torque settings

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Timing chain tensioner

Timing chain tensioner (Lower) 10 (Nm)

Timing chain tensioner (Upper) 70 (Nm)

Renew the seal (Threads lightly oiled)

Timing chain tensioner guide

Timing chain tensioner guide (Renew the bolts) 20 (Nm)

Timing chain guide

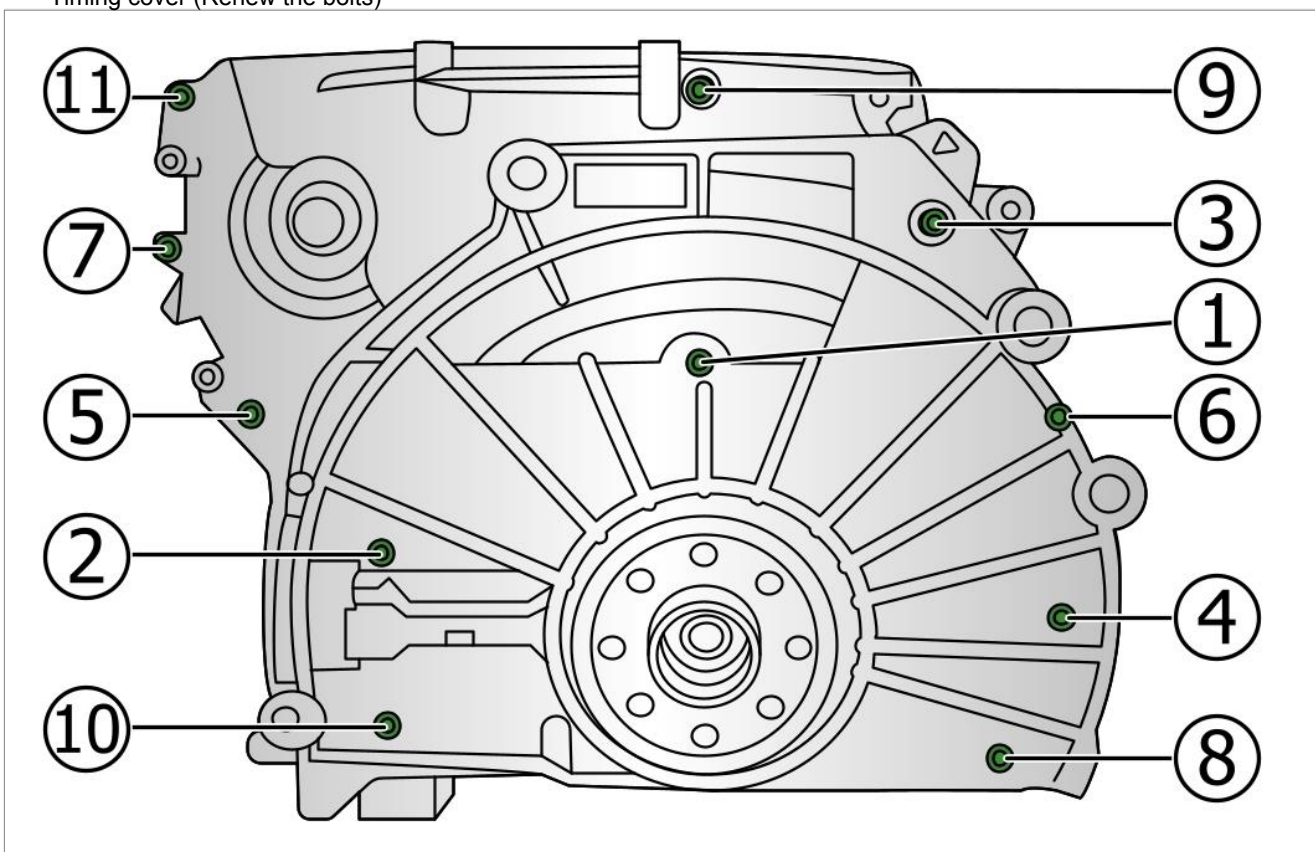
Timing chain guide (Renew the bolts) 20 (Nm)

High-pressure pump gearwheel

High-pressure pump gearwheel (Renew the cap and O-ring) 65 (Nm)

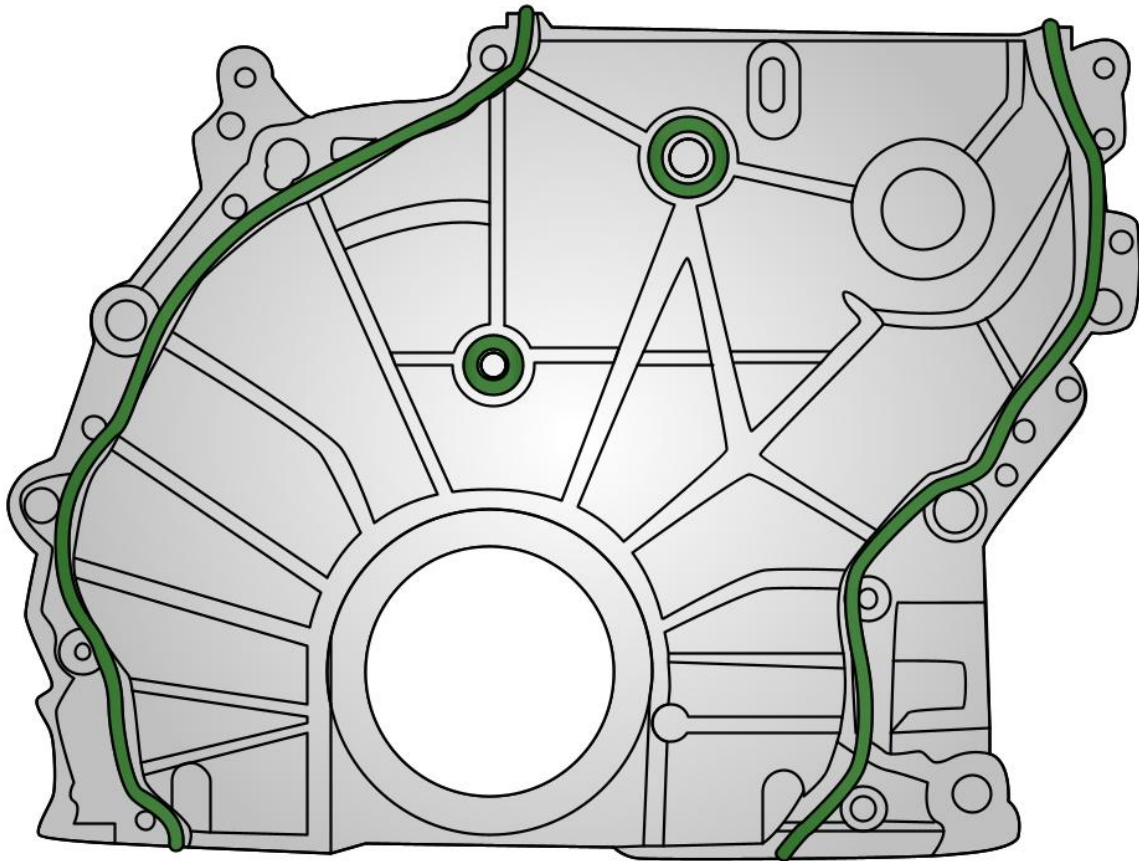
Timing cover

Timing cover (Renew the bolts)



Clean the contact surfaces

Apply sealant (BMW 83 19 0 404 517; Loctite 5970) 1.5 - 2.0 (mm)

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The component must be fitted within 12 minutes of applying the sealant

Stage 1 8 (Nm)

Stage 2 90 (°)

Oil pump

Oil pump (Renew the bolts)

Clean the contact surfaces

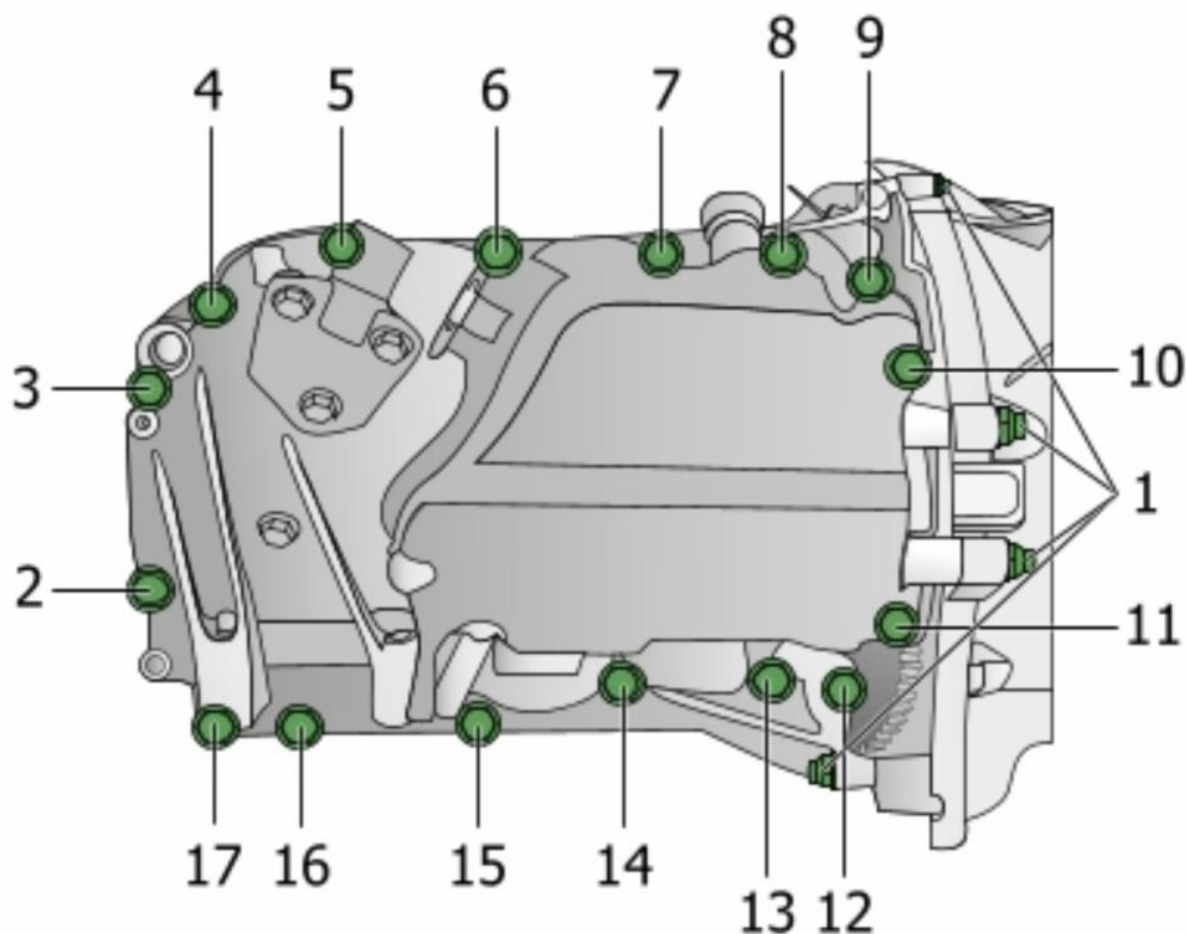
Stage 1 15 (Nm)

Stage 2 45 (°)

Oil sump

Sump (Clean the contact surfaces)

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Apply sealant 2.0 - 2.5 (mm)

Stage 1 (Tighten the bolts until full surface contact is made)

Stage 2 (Bolts 1) 5 (Nm)

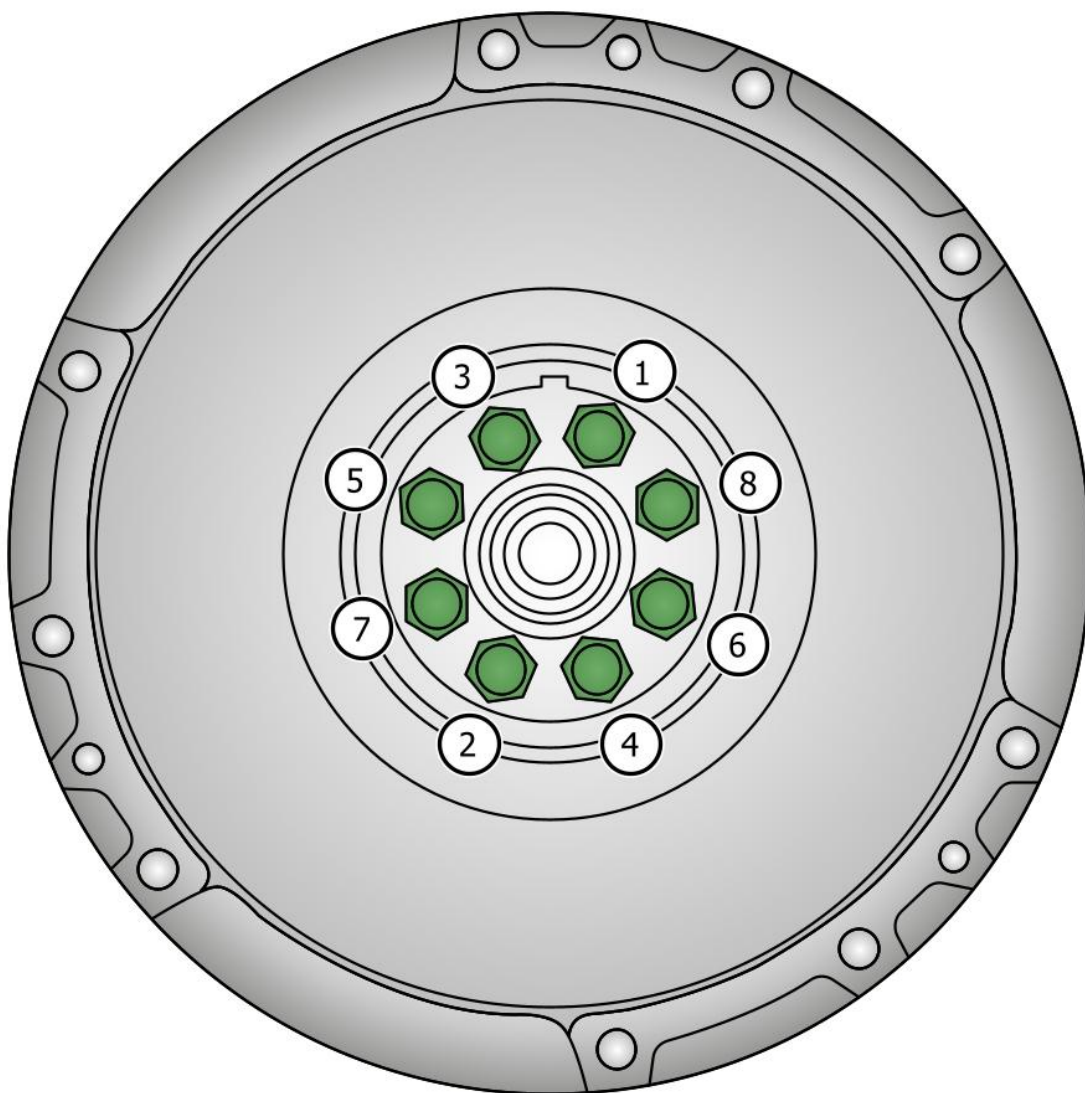
Stage 3 (Bolts 2 - 17) 24 (Nm)

Stage 4 (Bolts 1) 19 (Nm)

Flywheel

Flywheel

Renew the bolts

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Stage 1 45 (Nm)

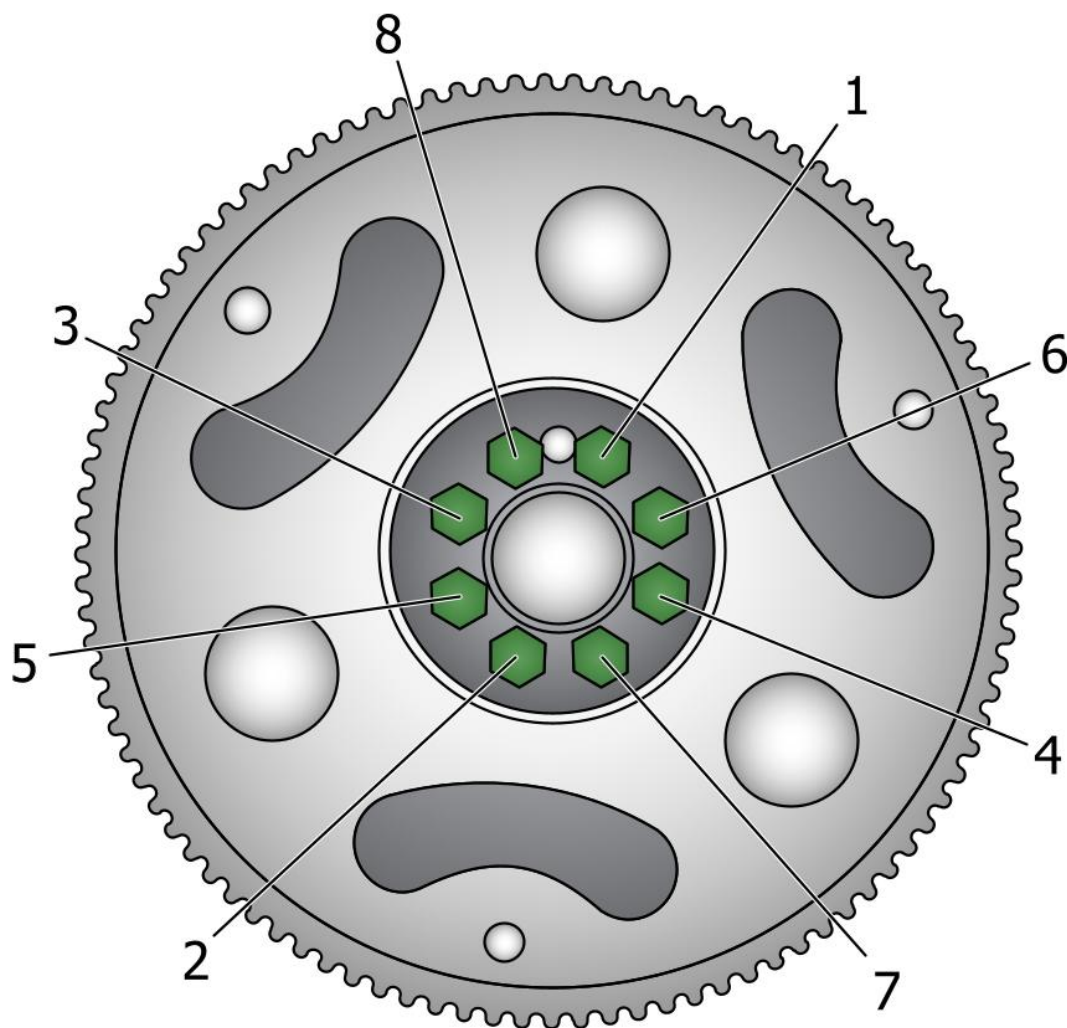
Stage 2 45 (°)

Flexplate

Flexplate

Renew the bolts

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Stage 1 45 (Nm)

Stage 2 45 (°)

Air-conditioning compressor

Air-conditioning compressor 38 (Nm)

Alternator

Alternator 38 (Nm)

Gearbox to engine:

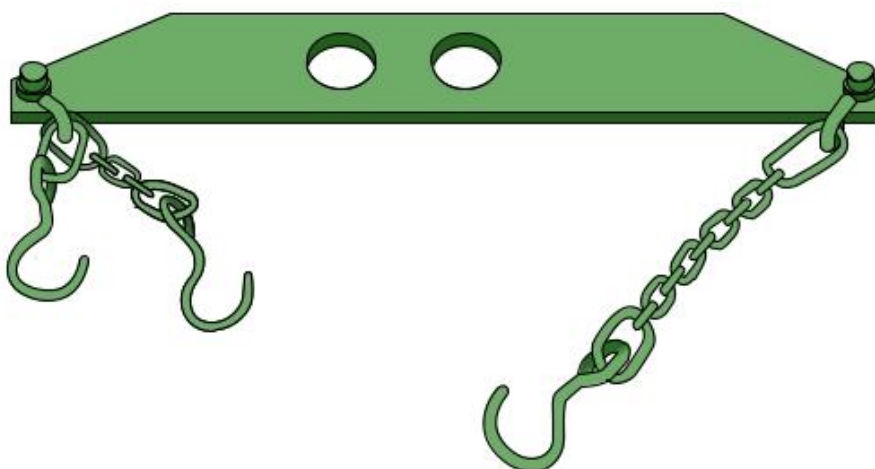
M12: (66 Nm)

M8: (19 Nm)

Transfer box to gearbox: (56 Nm)

Special tools

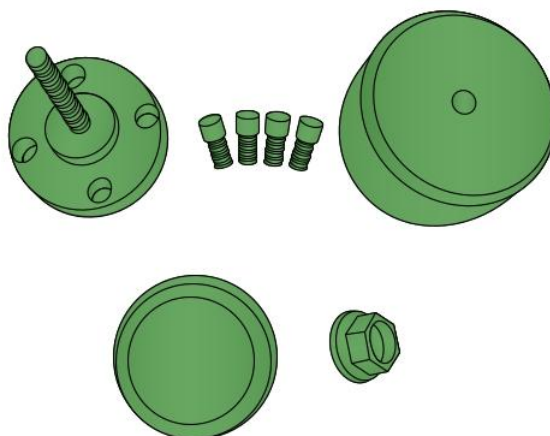
Engine hoist: (11 0 020)

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Socket: (11 6 480)

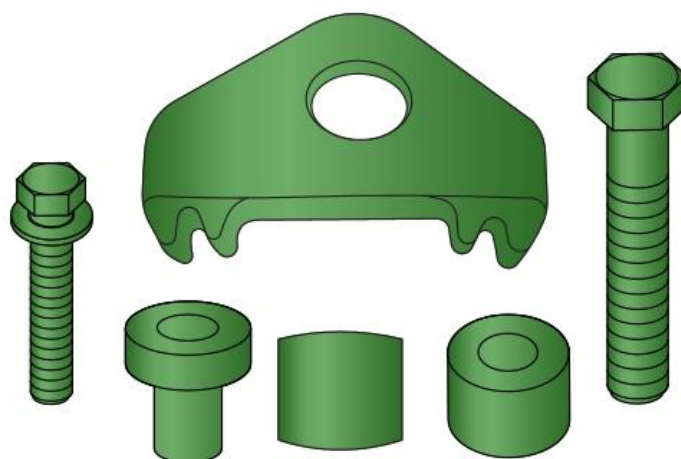


Oil seal fitting tool: (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)

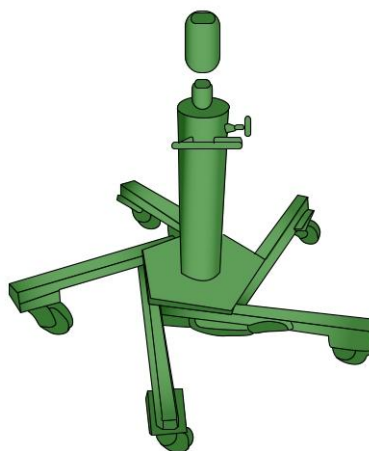


Flywheel locking tool: (11 9 260)

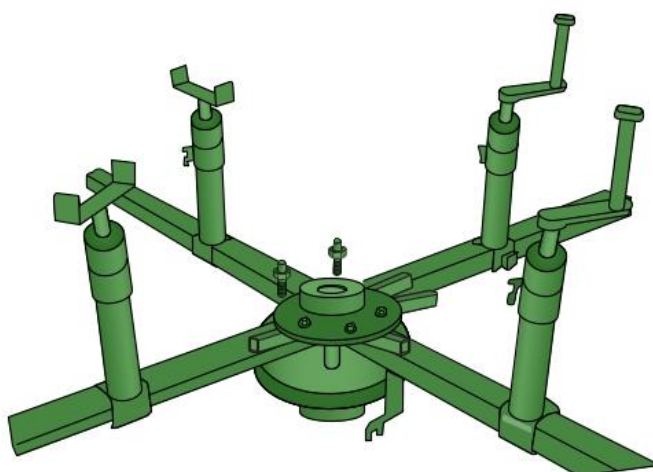
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Transmission jack: (00 2 030)

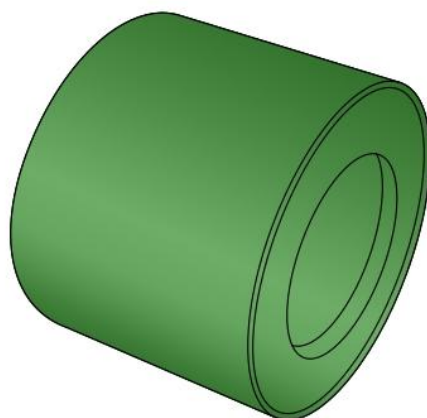


Subframe support: (31 5 250)

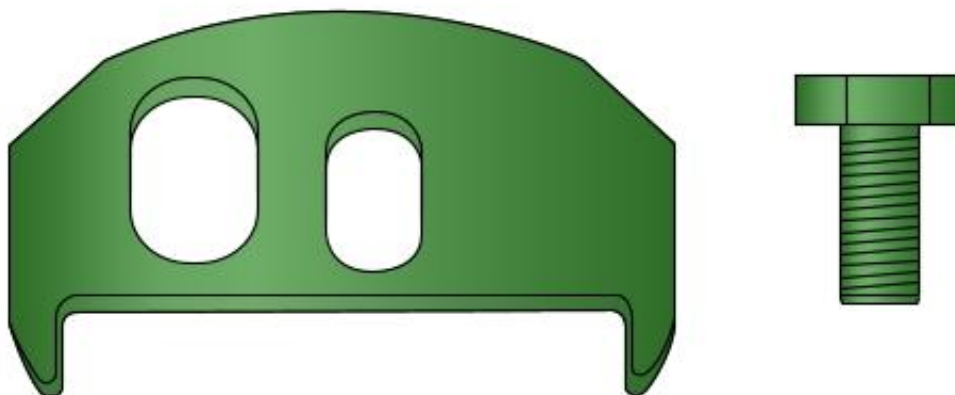


Bush: (11 9 265)

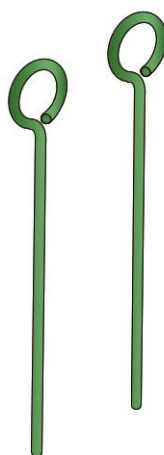
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Flywheel locking tool: (11 9 811)

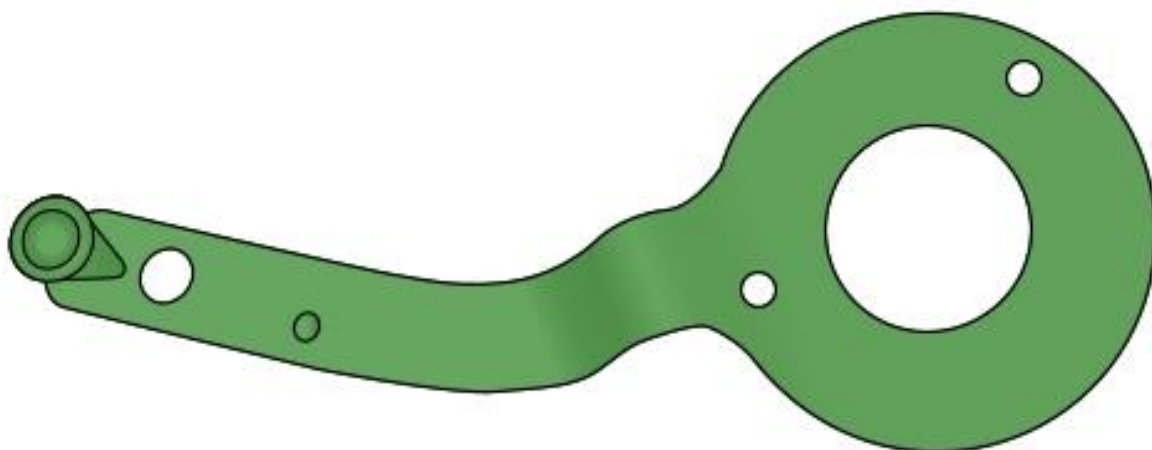


Locking pin: (11 4 120)

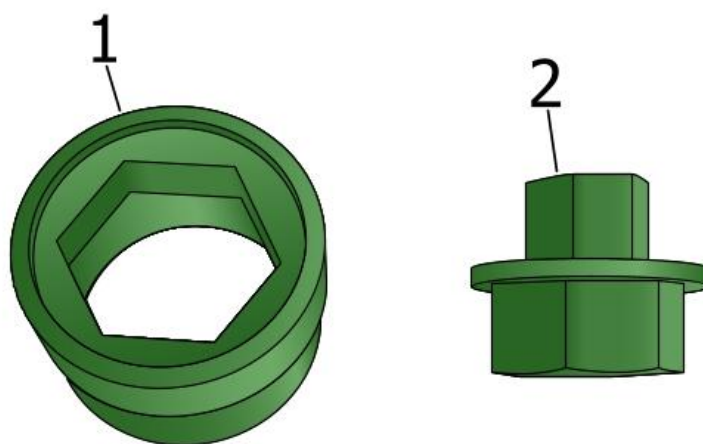


Crankshaft locking tool: (2 357 904)

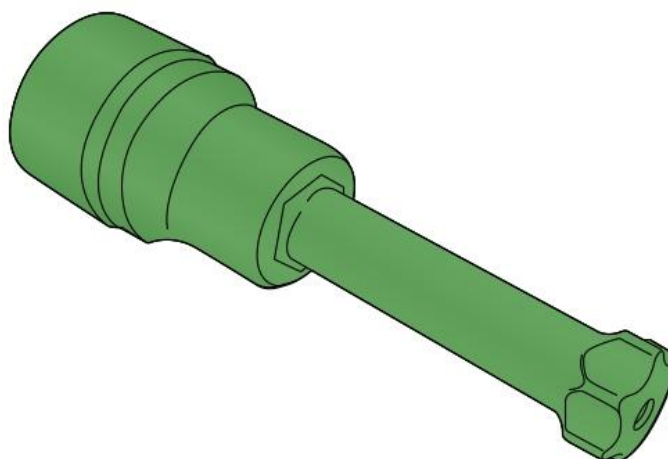
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Extractor: (11 8 030)

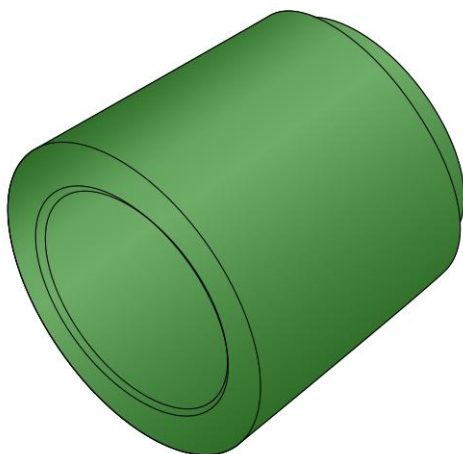


Socket: (11 4 180)

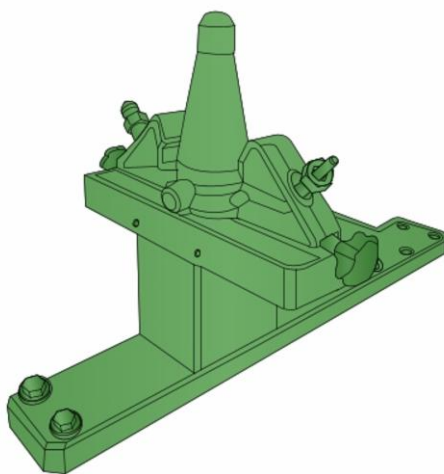


Oil seal fitting tool: (11 8 815)

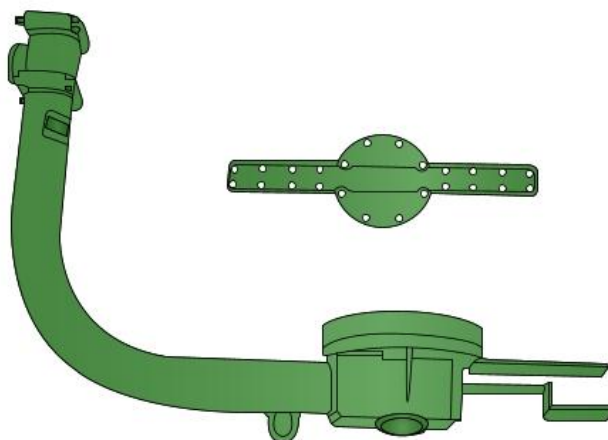
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Engine support: (2 413 082)

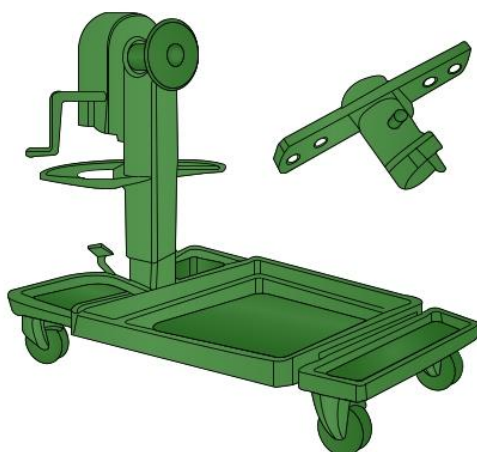


Engine support: (00 2 320)

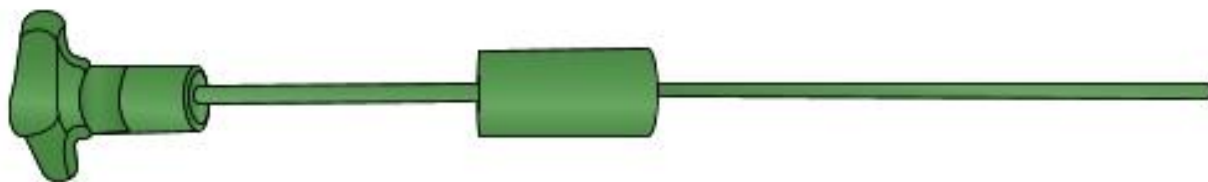


Engine support: (00 2 300)

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Oil seal removal tool: (23 0 490)

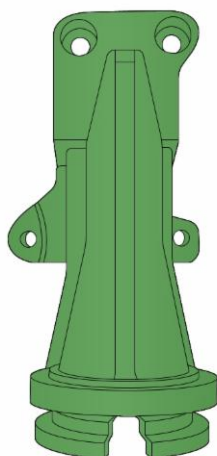


Subframe support: (2 184 136)

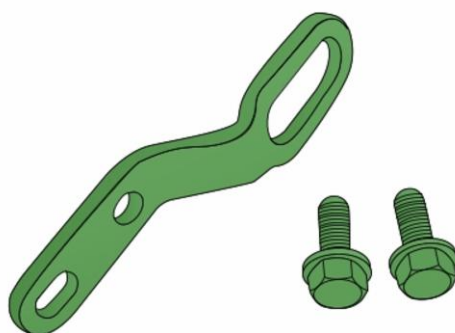


Engine support tool: (2 455 010)

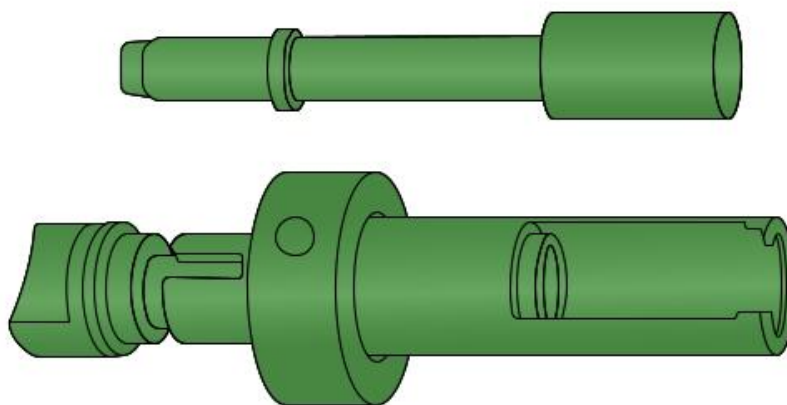
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Lifting eye: (2 459 012)



Blanking plug: (13 5 161, 13 5 162)

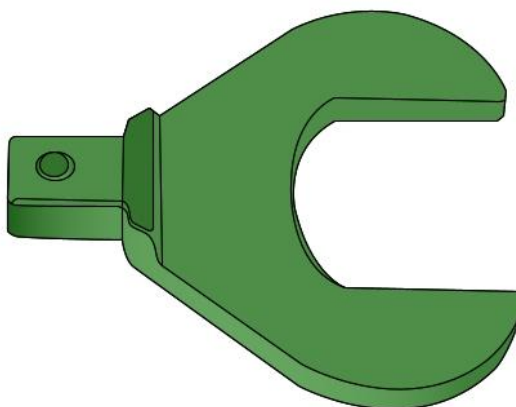


Engine lift: (2 220 718)

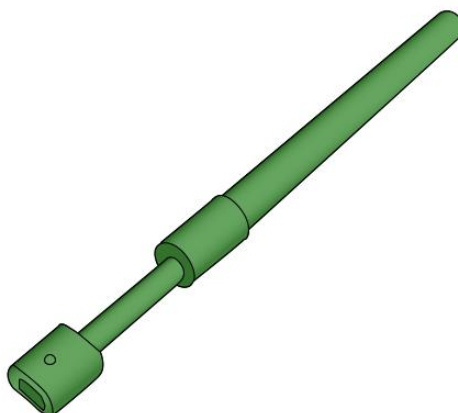
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Wrench: (33 0 080)

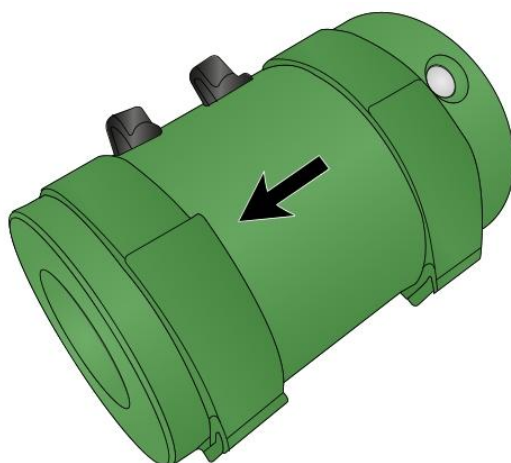


Wrench: (33 5 070)

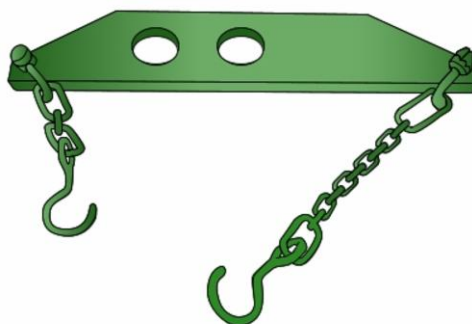


Protective sleeve: (2 407 851)

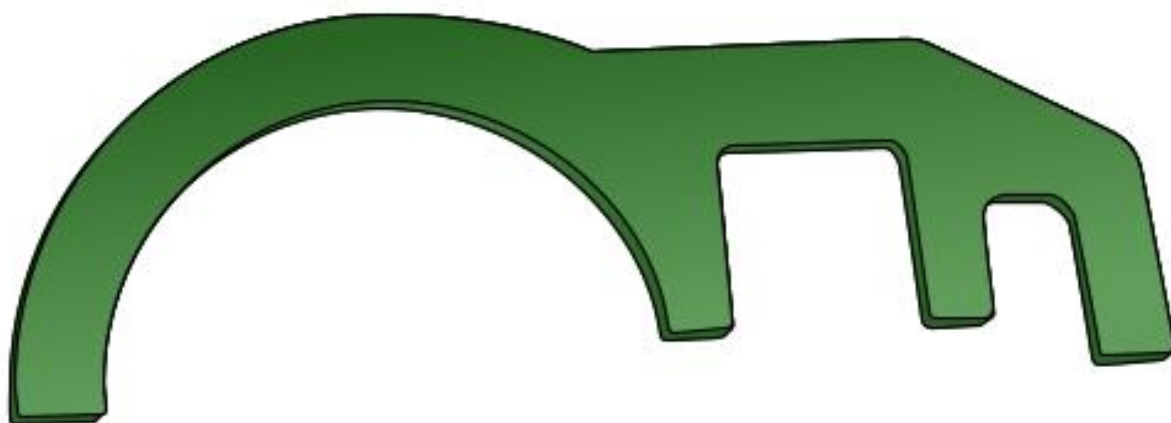
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Engine support tool: (11 0 000)



Camshaft blocking tool: (11 8 760)

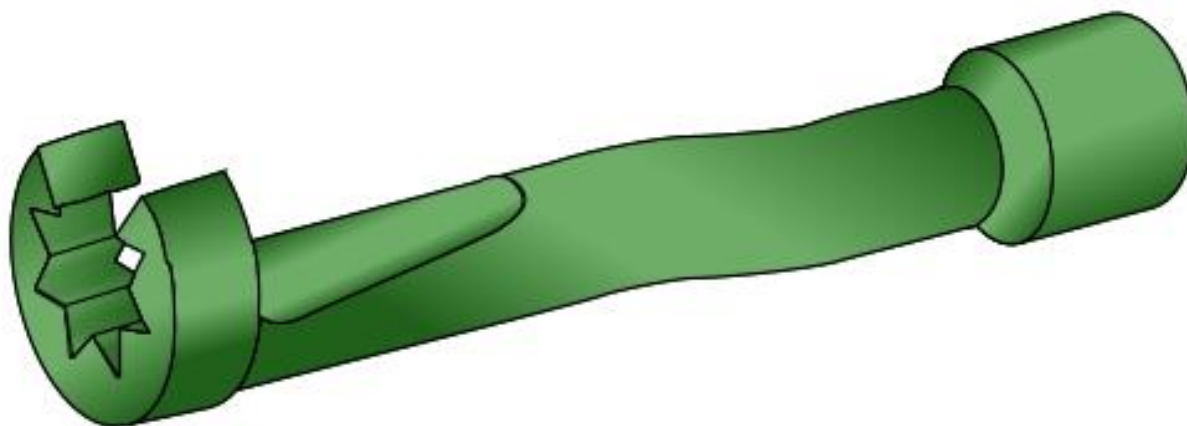


Flexplate locking tool: (2 288 380)

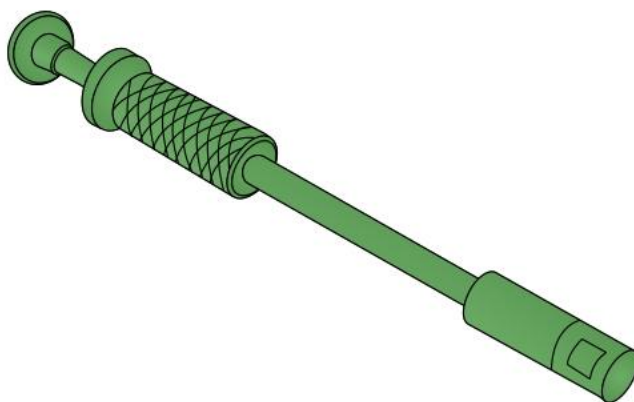
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Socket wrench: (37 1 152)



Injector removal tool: (13 5 250)

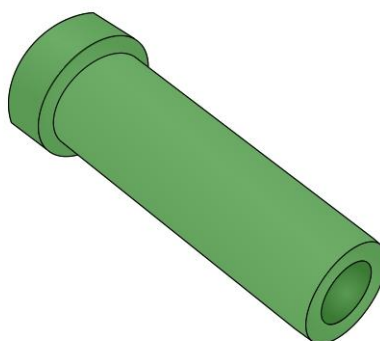


Injector removal tool: (13 5 252)

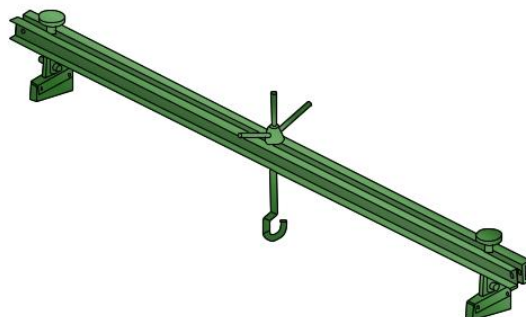
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Injector removal tool: (13 5 253)

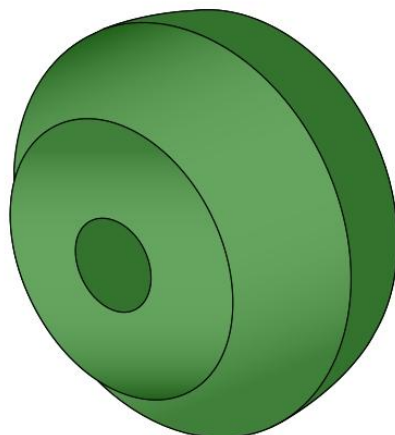


Engine support: (2 414 691)

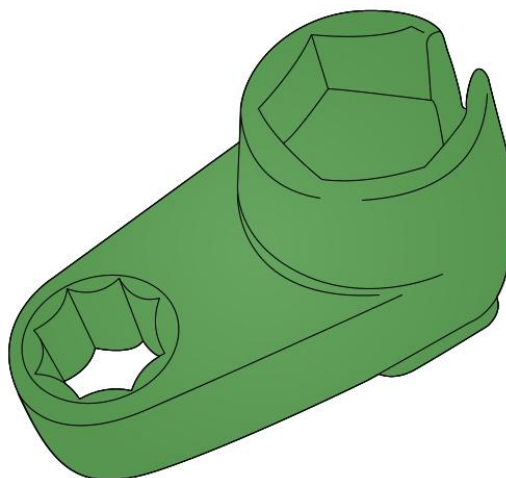


Engine support: (2 357 071)

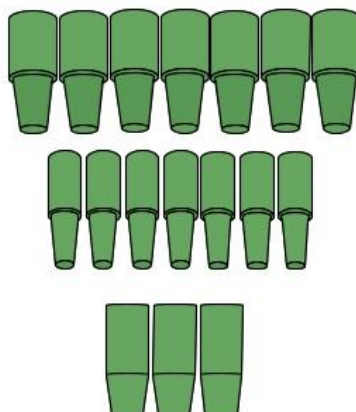
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Socket wrench: (11 4 260)

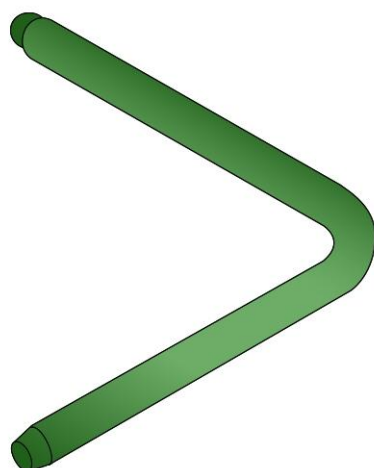


Blanking plugs: (2 364 711)

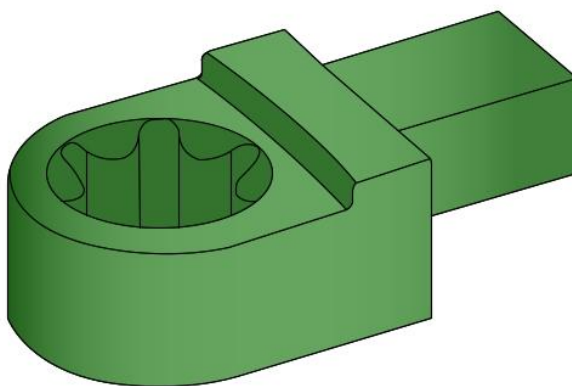


Tensioner locking pin: (11 0 390)

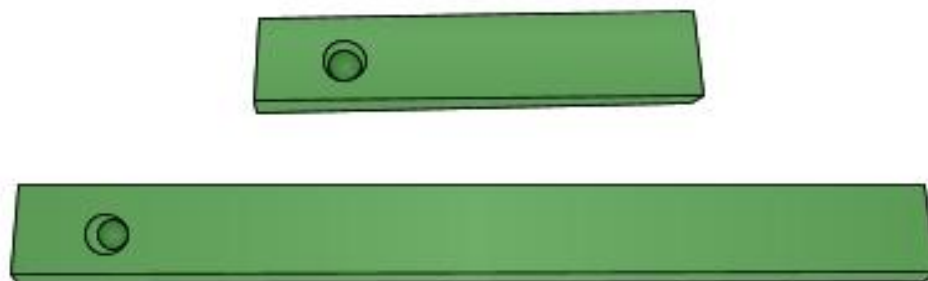
Repair Manuals



Special tool: (AF 16)



Special tools: (11 8 780)



Clutch compressing tool: (21 2 170)

Repair Manuals

